Crop Recommendation System

Using Machine Learning

Project Report

Fundamentals of AI and ML — Evaluated Course (BYOP)

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1. Introduction

Agriculture is the backbone of India's economy, employing nearly 50% of the workforce. Despite this, farmers — especially small and marginal ones — often make crop selection decisions based on tradition or guesswork rather than data. This leads to poor yields, resource wastage, and financial losses. Madhya Pradesh, India's largest state by area, is predominantly agrarian, making this problem especially relevant.

This project applies Machine Learning to recommend the most suitable crop to cultivate, given measurable soil and climate conditions. By learning patterns from historical data, the model bridges the gap between soil science and practical farming decisions.

2. Problem Statement

Given measurable inputs about a field and its local climate — nitrogen (N), phosphorus (P), potassium (K) levels in the soil, temperature, humidity, pH, and annual rainfall — predict the most suitable crop for cultivation.

This is a multi-class classification problem with 10 possible output classes (crops).

3. Why This Problem Matters

- Crop mismatch is a leading cause of crop failure in rain-fed agriculture.
- Soil testing is now accessible to farmers but interpretation remains a challenge.
- A data-driven recommendation tool can bridge the gap between soil data and farming action.
- Scalable: the same approach works for any region with appropriate training data.
- Directly aligned with India's Digital Agriculture Mission.

4. Dataset

4.1 Description

The dataset used in this project is based on the publicly available Crop Recommendation Dataset (Kaggle). For this project, a representative version was programmatically generated using domain-realistic ranges for each crop, resulting in 1,100 samples (110 per crop).

4.2 Input Features

Feature	Description	Unit	Example
N	Nitrogen content in soil	mg/kg	90
P	Phosphorus content in soil	mg/kg	42
K	Potassium content in soil	mg/kg	43
temperature	Average ambient temperature	°C	21.0
humidity	Relative humidity	%	82.0
ph	Soil pH value	0–14	6.5
rainfall	Annual rainfall	mm	202.0

label	Crop name (target variable)	—	rice

4.3 Crop Classes

Crop	Key Growing Conditions
Rice	High humidity (70–90%), high rainfall (150–250 mm)
Wheat	Moderate temperature (15–25°C), low rainfall
Maize	Warm climate, moderate rainfall
Cotton	High temperature, low rainfall, high N
Sugarcane	Warm, high N, good rainfall
Mung Bean	Warm, moderate humidity, medium N
Lentil	Cool, dry, very low N
Pomegranate	Hot and dry, low N
Mango	Tropical, warm climate
Banana	Humid, high N, P and K

5. Approach & Methodology

5.1 Algorithm: Random Forest Classifier

Random Forest is an ensemble learning algorithm that builds multiple Decision Trees during training and merges their outputs. It was chosen because:

- It handles non-linear relationships between features and labels well.
- It is robust to overfitting compared to a single Decision Tree.
- It provides feature importance scores — directly useful for agricultural insight.
- It requires minimal hyperparameter tuning and works well out of the box.

5.2 ML Pipeline

Step	Action	Tool / Method
1	Data Generation	NumPy random sampling within crop-specific ranges
2	Exploratory Analysis	pandas describe(), value_counts()
3	Label Encoding	sklearn LabelEncoder
4	Train/Test Split	80% train, 20% test (stratified)
5	Model Training	RandomForestClassifier (100 estimators)
6	Evaluation	accuracy_score, classification_report
7	Feature Importance	model.feature_importances_
8	Prediction Demo	model.predict() on new sample inputs

6. Results

6.1 Model Performance

Metric	Value

Overall Accuracy	97.73%
Macro Avg Precision	0.98
Macro Avg Recall	0.98
Macro Avg F1-Score	0.98
Test Set Size	220 samples (20% of 1,100)

6.2 Feature Importance

Rank	Feature	Importance	Agricultural Significance
1	N (Nitrogen)	0.2294	Primary nutrient; varies sharply by crop family
2	Rainfall	0.1960	Water availability drives crop suitability
3	P (Phosphorus)	0.1678	Root development; varies by crop family
4	Humidity	0.1433	Disease risk and transpiration rate
5	Temperature	0.1191	Growth stage timing
6	K (Potassium)	0.1051	Stress resistance and fruit quality
7	pH	0.0392	Nutrient availability in soil

6.3 Sample Predictions

N	P	K	Temp	Humidity	pH	Rainfall	Predicted Crop
90	42	43	21°C	82%	6.5	202 mm	RICE ✓
100	55	60	28°C	62%	7.0	80 mm	COTTON ✓
25	60	38	20°C	55%	6.8	45 mm	LENTIL ✓

7. Key Decisions & Justifications

- Random Forest over Logistic Regression: Crop decision boundaries are non-linear (e.g., rice and banana both need high humidity but different N levels). Random Forest handles this better.
- Stratified Split: Ensures each crop class is proportionally represented in train and test, preventing evaluation bias.
- Feature Importance Analysis: Added to make the model interpretable for a farming context — not just 'what to grow' but 'why this field suits this crop'.
- Domain-Realistic Data: Each crop's data was generated within agronomically correct ranges, grounding the project in real-world science.

8. Challenges Faced

- Understanding which features matter in agriculture required reading about soil science and how N, P, K levels relate to different crop families.
- Balancing simplicity and completeness — the final code is clean and readable while covering a full ML pipeline.
- Avoiding overfitting — solved using stratified splits, ensemble method (100 trees), and careful evaluation on unseen test data.

9. What I Learned

- The complete supervised learning pipeline: data preparation → model training → evaluation → interpretation.
- How Random Forest works: an ensemble of Decision Trees trained on random subsets of data and features, voting for the final output.
- Feature importance: ML models are not just black boxes — their outputs can be interpreted to generate real-world insight.
- How to frame a real-world problem as a classification task and map it to the concepts taught in this course.
- Evaluation metrics beyond accuracy — precision, recall, and F1-score matter especially when class distributions could be imbalanced.

10. Future Scope

- Integrate real soil test APIs or government agriculture databases for live input.
- Build a simple web interface (Flask/Streamlit) for farmer-friendly usage.
- Expand to 22+ crops using the full Kaggle dataset.
- Add regional filtering (e.g., crops suitable specifically for Madhya Pradesh).
- Compare models: Logistic Regression vs Decision Tree vs Random Forest vs SVM.

11. Conclusion

This project demonstrates how a fundamental Machine Learning concept — multi-class classification using Random Forest — can be applied to a real-world agricultural problem. The model achieves 97.73% accuracy and produces interpretable feature importance scores, making it both technically sound and practically meaningful.

The problem of crop selection is not hypothetical: it affects millions of farmers in India every season. While this project is a learning exercise, the approach it demonstrates — soil and climate data feeding into an ML model to produce a crop recommendation — is the same approach used by real agricultural AI startups. This makes it both a meaningful application of course concepts and a solid foundation for future development.

12. References

- Scikit-learn Documentation — https://scikit-learn.org
- Crop Recommendation Dataset — Kaggle (Atharva Ingle, 2020)
- Introduction to Machine Learning with Python — Müller & Guido (O'Reilly)
- Indian Council of Agricultural Research (ICAR) — Crop nutrient requirements
- Fundamentals of AI and ML — Course Materials, VIT